What is your experience with C?

CS 340

C without the ++ (0b01)

What percentage of the class do you think has successfully written a program in C?

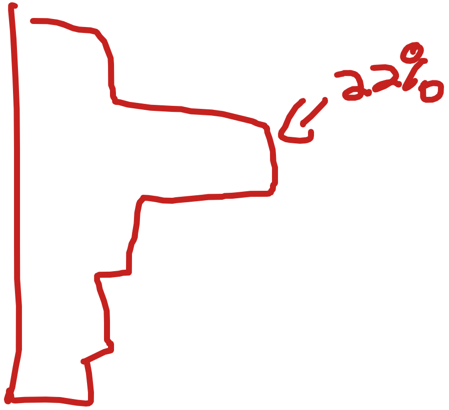
clicker.cs.illinois.edu

Q2

~Code~
340

21% have written a program in C

22% winner



Updates

1. MP 1.1, MP 1.2 → sept 15

2. Environment check-off → 1%

3. HW 1 → due tomorrow midnight

4. Collaboration time/office hours today!

   – 4:00   4:30 – 5:00

C without the C++

LGs:

- Be able to read and understand C code
- Be able to write C code from scratch

1. Review
2. Why C
3. Memory, Arrays, and printf()
4. Dynamic memory
5. Using arrays
6. c-strings

Review: What could print on lines 7 and 8?

```
4   int main(){
5       int *arr = new int[4];
6       cout << arr << endl;
7       cout << &arr << endl;
8       cout << *arr << endl;
9       delete[] arr;
10  }
```

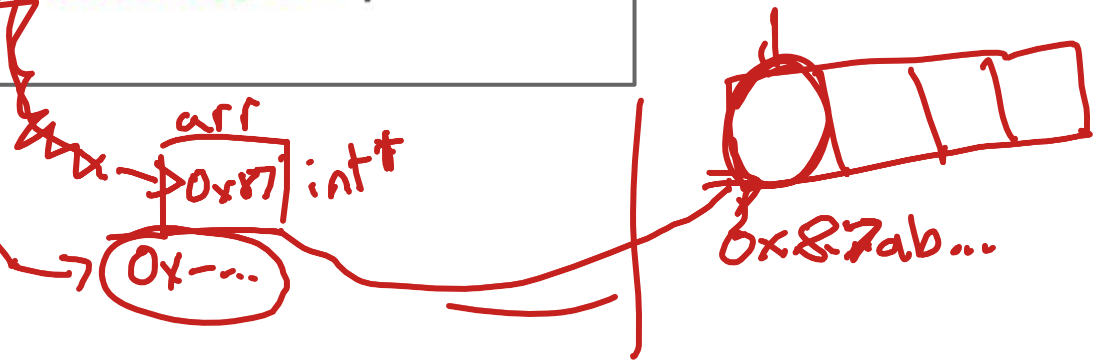
clicker.cs.illinois.edu

Q3

~Code~
340

0x87ab39012a

0x4930... 0



Review: Which is a separate computer (hardware, OS, etc...) from your local machine?

clicker.cs.illinois.edu

Q4

~Code~
340

a) Docker
b) Terminal/Shell
c) Virtual Machine (VM)

Why do we use a VM in CS 340? Multiple select available.

clicker.cs.illinois.edu

Q5

~Code~

340

a) So all the MPs can be consistent across all student computers
b) So specific MPs can be timed more precisely
c) So we can connect student computers for the final project

But why C?

Lower-level... closer to the hardware

Used in fields such as...

networking, security, OS

Used in...

future courses

OS kernel code

How is C different from C++?

1. No classes
2. No templates
3. No function or operator overloading
4. No new or delete
5. No pass-by-reference
6. No standard C++ library

vectors, strings, lists,

How confused/nervous are you to code in C without the ++ features?

Memory, Arrays, and printf()

Data Stored as Bytes

Physically, computer hardware...

1 or 0 bit

To make binary more practical we...

byte == 8 bits

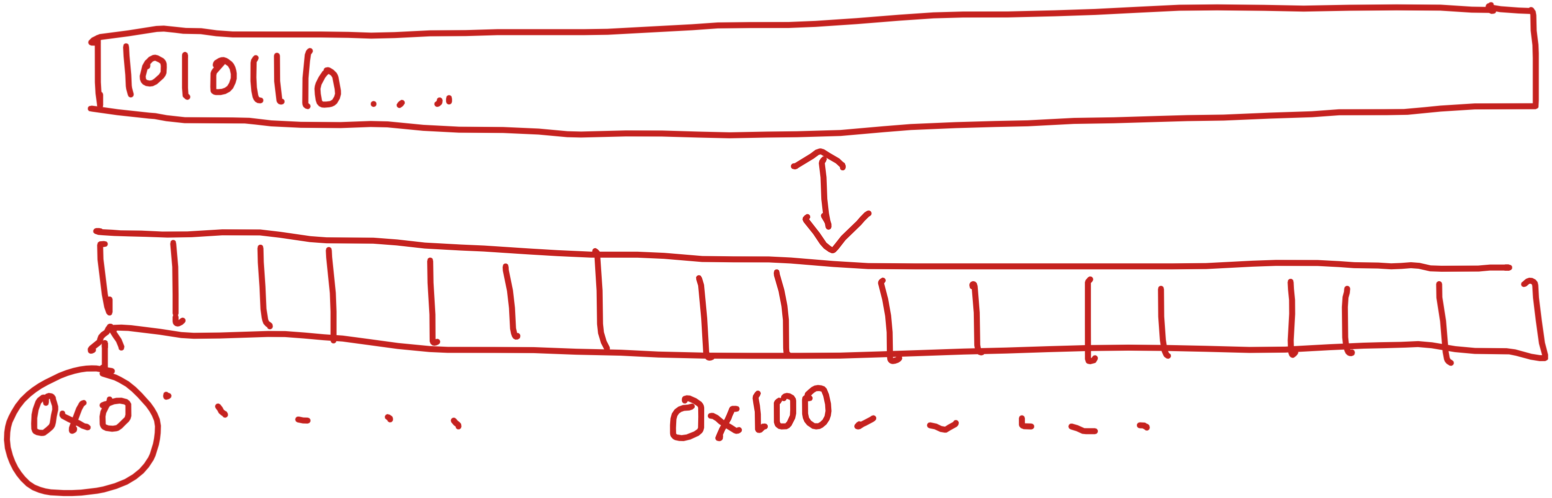
in hex 0xFa326

Data types have a size (in bytes)...

int = 4 bytes 32 bits

char = 1 bytes 8 bits

Addresses in a C/C++ Program



segfault → access an address you should not

What is the difference between an array and a vector?

clicker.cs.illinois.edu

Q6

~Code~
340

What is an array?

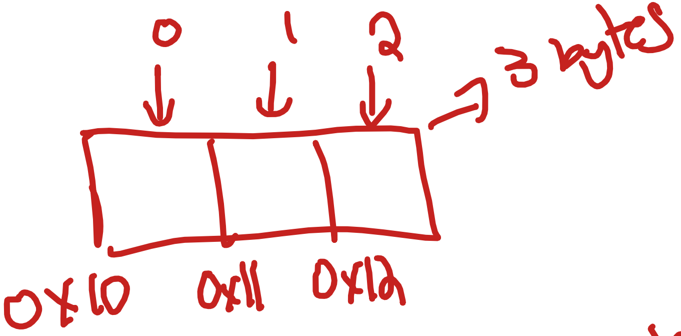
contiguous memory

char arr[3];

1 byte

int arr[2];

4 bytes



Would line 5 or 6 compile?

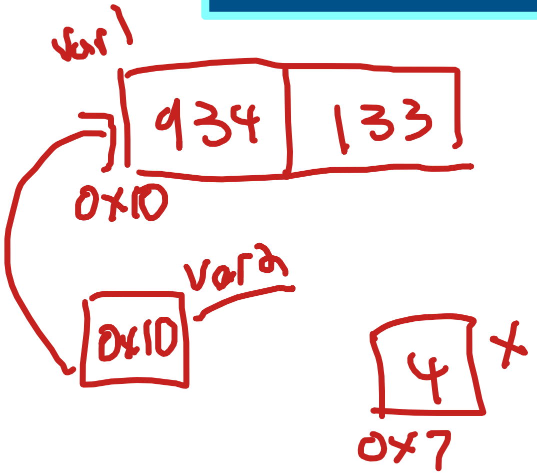
clicker.cs.illinois.edu

Q7

~Code~
340

[ ] index

```
1   int main(){
2       int var1[2] = {934, 133};
3       int *var2 = &var1[0];
4       int x = 4;
5       //var2 = &x;
6       //var1 = &x;
7   }
```

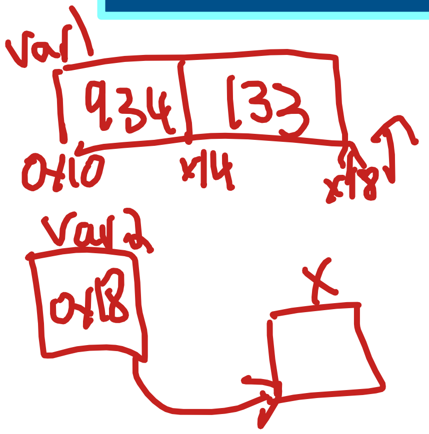


Challenge: Would this code segfault?

var1[2]

*(var1 + 2)

```
int main(){
    int var1[2] = {934, 133};
    int *var2 = &var1[2];
    int x = 4;
    var2 = &x;
}
```



clicker.cs.illinois.edu

Q8

~Code~

340

Pointer Math

navigate memory

- arrays
- files

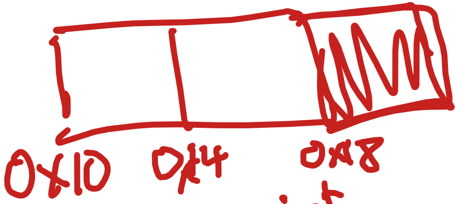
address + int value == address + (sizeof) * (int value)

int arr[3];

arr + 2;

address

4



int

0x10 + 2(4) = 0x18

Arrays with pointer math

```
int arr[3] = {6, 7, 8};
```

~~arr[~~ arr[1]

*(arr+1)

7

any number

int printf(const char * format, args);

- %d or %i: for integers
- %#x: for 0xHEX_NUMBER
- %c: for characters
- %s: for C-strings
- ...more

```
int main() {
    int x = 5;
    int *ptr = &x;
    char *str = "hi";
    printf("will print: %i, %#x, %s", x, ptr, str);
}
```

input

```
will print: 5, 0xd88ae92c, hi
```

Would these print the same or different things?

clicker.cs.illinois.edu

Q9

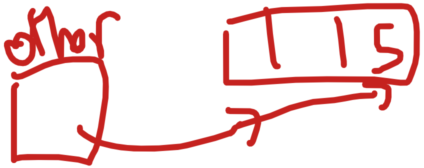
~Code~

340

Same

```c
int main() {
    int arr[3] = {1, 4, 5};
    int *other = arr + 2;
    printf("%d\n", *other);
}
```

=5




```c
int main() {
    int arr[3] = {1, 4, 5};
    int *other = arr + 2;
    printf("%d\n", arr[2]);
}
```

=5

If line 17 prints "0x13000", what would line 19 print?

clicker.cs.illinois.edu

Q10

~Code~
340

```
15  int main() {
16      int arr[3] = {1, 4, 5};
17      printf("%#x\n", arr);
18      int *other = arr + 2;
19      printf("%#x\n", other);
20  }
```

ints

+8

0x1308

This compiles! What do you think would print?

clicker.cs.illinois.edu

Q11

~Code~

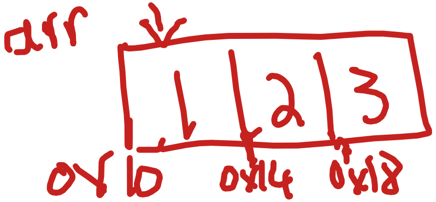
340

```c
int main() {
    int arr[3] = {1, 4, 5};
    int *other = arr + 2;
    printf("%d\n", other[-1]);
}
```

&(arr[2]

*(other + -1)

Explain to your neighbor: why do you think we index starting at 0 in C/C++ when accessing a vector or array?

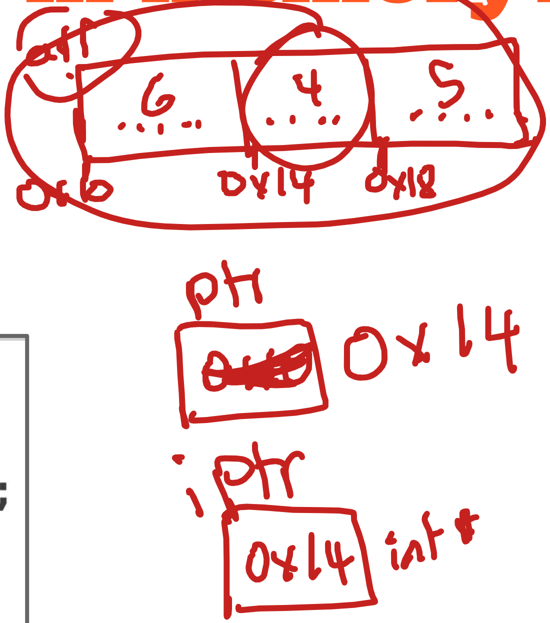


*(arr + 0) → 1

*(arr + 1) → 2

How does the computer know what type something is in memory?

```c
int main() {
    int arr[3] = {6, 4, 5};
    char *ptr = (char*)(arr);
    ptr = ptr + 4;
    int *iptr = (int*)(ptr);
    printf("%d\n", *iptr);
}
```



Big Ideas

1. Pointer math increases a pointer by multiples of sizeof(*ptr).

2. Arrays are contiguous bytes that can be accessed through pointer math.

3. Bytes are bytes… the type of the variable determines how the bytes are interpreted.

Reading Files in C

A file is made up of....

info .... bytes .... 1's and 0's

Relevant Functions

FILE * fopen(const char * file, const char* mode)

size_t fread(void *restrict ptr, size_t size, size_t nitems, FILE *restrict stream);

```
int main(){
    FILE *fl = fopen("file.txt", "r");
    int x;
    int read = fread(&x, 4, 1, fl);
    fclose(fl);
}
```

How would you fix this code?

clicker.cs.illinois.edu

Q12

~Code~
340

```c
#include <stdio.h>

int main(){
    FILE *fl = fopen("file.txt", 'r');
    int x;
    int read = fread(&x, 4, 1, fl);
    fclose(fl);
}
```

input | stderr

Compilation failed due to following error(s).

```
main.c: In function ‘main’:
main.c:4:34: error: passing argument 2 of ‘fopen’ makes pointer from integer without a cast
    4 |     FILE *fl = fopen("file.txt", 'r');
      |                                  ^~~
      |                                  |
      |                                  int
      |
```

How would I read 3 ints into the array called x?

a×b=12

x ; a;b : fl

or

&x

clicker.cs.illinois.edu

Q13

~Code~
340

```c
int main(){
    FILE *fl = fopen("file.txt", "r");
    int x[3];
    int read = fread(               );
    fclose(fl);
}
```

Relevant Functions

“5 6 $\boxed{7\ 8}$ 9” file

bytes

int fseek(FILE *stream, long int offset, int whence);

SEEK_SET - start

SEEK_CUR - cur

SEEK_END - end

x

5	7	8

```
int main(){
    FILE *fl = fopen("file.txt", "r");
    int x[3];
    int read = fread(x, 4, 1, fl);
    fseek(fl, 8, SEEK_SET);
    int read = fread(x+1, 4, 2, fl);
    fclose(fl);
}
```